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RESEARCH-ARTICLE

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Enabling Granularity-customizable Geocast in Network Layer Using P4-based Software Defined Network

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ABSTRACT

Geocast is a one-to-many communication paradigm for sending the data packets to a designated area rather than an IP address. However, the traditional IP-based solutions cannot cope with the challenges in flexibility, mobility, and implementation overhead in situations where users are requiring a granularity-customizable geocast. Therefore, we propose to implement the granularity-customizable geocast at the edge of the network by embedding a novel network-layer addressing scheme using P4-based Software Defined Network (SDN). In this paper, a geographic subdivision and coding method named EMD is used to subdivide the global region. Then, we introduce the GeoIP as the network address and propose a GeoIP packet processing scheme by utilizing P4. The proposed GeoIP-based addressing scheme is deployed at the edge of the network. To make our design compatible with the current Internet, a Geolocation Name Service (GNS) system is designed to support the Internet-wide geocast. In addition, a prototype system is built to implement and evaluate our design. Experiment results show that the proposed design is feasible to provide granularity-customizable geocast at a relatively low cost.

CCS CONCEPTS

• **Networks** → *Naming and addressing; Network architectures.*

KEYWORDS

Geocast, Network layer addressing, SDN, P4

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1 INTRODUCTION

The developing Internet technologies indicate some foreseeable future network use cases like the geolocation-oriented commanding on the battlefield or the vehicle scheduling for autonomous driving. In these use cases, users may have the requirement of sending data

packets to the designated area of customized granularity. Geocast is a one-to-many communication paradigm for sending data packets to the end devices in a designated area instead of an end device with a specific IP address[1]. It is effective in situations where a large number of nodes within a geographical area need to be reached[2]. Because the end devices receive data packets based on their location rather than a subscription model. Geocast is a potential solution to meet the above requirements.

The current Internet provides some IP-based approaches to support geocast which predominantly rely on specialized location-based services[3, 4]. For instance, users could query the Geolocation-to-IP database (e.g., Akamai, Quova, Maxmind, Geobytes, Digital Envoy, IPInfoDB[5]) to obtain the IP addresses of the end devices in a designated area. Then, users could either send data packets to these end devices or send data packets to the specific wireless Access Points (APs) or Road Side Units (RSUs) to implement geocast for the end devices within their signal range[6].

However, the above IP-based approaches face three main challenges. In terms of flexibility, the granularity of the designated area completely depends on the signal range of AP/RSU. It is difficult to implement a fine-grained geocast when the designated area is smaller than the signal range of AP/RSU[7, 8]. In terms of overhead, sending packets to the specific IP addresses has the obvious demerit of sending one packet per destination, incurring additional communication overhead when numerous devices are present in the designated area[9, 10]. In terms of mobility, the cost for a Geolocation-to-IP database to update its mapping entries will be unacceptable in situations where the receivers have a high rate of position change[11, 12].

Some other studies about location-based routing protocols have been conducted, attempting to implement geocast in the network layer through designing a novel routing scheme based on geographic addressing and forwarding, e.g., the GeoTORA[13], Geo-AKOM[14], DA-GSAF[15], EECSG[16], SMGSAF[17], GeoNetworking[18], G-locON[19], etc.

Nevertheless, the above routing approaches are mostly designed for wireless scenarios. The connectivity and channel usage are usually their primary concerns. Therefore, these routing approaches are usually inflexible to implement the granularity-customizable geocast. Moreover, the non-IP network-layer addressing makes these routing approaches rather incompatible with the current Internet.

In this paper, we propose a combined solution to enable granularity-customizable geocast for the current Internet: Implementing granularity-customizable geocast at the edge of the network through embedding a novel network-layer addressing scheme

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using the emerging P4-based SDN, then enhancing the compatibility with the assistance of mapping service. P4-based SDN provides a flexible data plane to customize the packet processing[20, 21], making it possible to embed a location-aware packet processing scheme in the network. Then, we could provide a mapping service on the control plane to make our design compatible with the current Internet. Our main contributions are summarized as follows:

- The global region is subdivided based on a geographic subdivision and coding method of EMD (Extended model based on the Mapping Division). Then we introduce the GeoIP as the name to identify the geographical areas of customized granularity.
- A GeoIP packet processing scheme is designed using the P4-based SDN to implement the granularity-customizable geocast at the edge of the network.
- A Geolocation Name Service (GNS) system is designed to make our design compatible with the current Internet by resolving a GeoIP to an IP address.
- The prototype system is built to implement and evaluate our design. Experiment results show that our design is feasible to provide granularity-customizable geocast at a relatively low cost.

The remainder of this paper is organized as follows. Section 2 presents the design goals and system model. Section 3 expounds on the detail of our solution design. Section 4 gives the implementation and evaluation. Section 5 presents concluding remarks.

2 DESIGN GOALS & SYSTEM MODEL

2.1 Design Goals

We present the design goals of implementing the granularity-customizable geocast as follows:

- (1) **Flexibility.** Users may require sending messages towards a customized fine-grained destination area when using geocast services. Therefore, it is important to provide a flexible network-layer naming method to identify the destination area of customized granularity.
- (2) **Mobility.** Geocast is usually applied in mobile scenarios, e.g., vehicle scheduling for autonomous driving. It is necessary to make network layer forwarding insensitive to the changes in the receiver's location.
- (3) **Compatibility.** Enabling an Internet-wide granularity-customizable geocast would be potentially valuable to improve the service capability of the Internet. Hence, enhancing the compatibility with the current Internet is also one of our design goals.

2.2 System Model Design

According to the proposed design goals, the system model of our design includes four parts and is illustrated as follows:

- (1) **EMD-based subdivision and coding:** We use the geomatics EMD subdivision and coding method to subdivide the global region into hierarchical areas. The subdivided area of any customized granularity is encoded with a unique binary code. Then we introduce the concept of the domain and

the subordinate subarea as the basis of implementing the granularity-customizable geocast.

- (2) **GeoIP-based addressing:** Based on the EMD coding method, we introduce GeoIP as the name of the subdivided area. A GeoIP is assigned to the AP of the subarea it represents and serves as the identifier for network layer packet forwarding. Because GeoIP is independent of the location of the end device, the network layer forwarding is insensitive to the changes in the receiver's location.
- (3) **GeoIP-based packet processing with P4:** A GeoIP packet processing scheme is proposed and embedded at the edge of the network by utilizing the P4 programmable data plane. Granularity-customizable geocast could be implemented through GeoIP packet forwarding and receiving.
- (4) **Geolocation Name Service:** To make our design compatible with the current IP network, we implement GeoIP using the IPv6 namespace and design a mapping service named Geolocation Name Service (GNS) on the control plane to resolve a GeoIP to an IP address.

3 SOLUTION DESIGN

3.1 EMD-based Geographic Subdivision & Coding

For flexibility concerns, our solution for granularity-customizable geocast is designed based on the EMD geographic mesh subdivision and coding method[22]. Firstly, the global region is subdivided hierarchically into multiple meshes by longitude and latitude interval. Secondly, binary coding is used to generate multilevel subdivision mesh code (SMC). Finally, every subdivided mesh of customized granularity is named by a unique SMC, the finer the granularity of a subdivided mesh is, the higher the level of its SMC is. The EMD-based mesh subdivision and coding method are shown in Fig. 1.

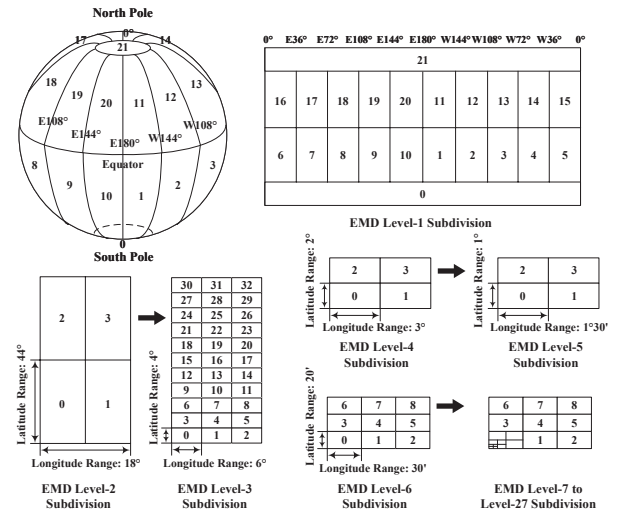


Figure 1: EMD mesh subdivision and coding method

SMC consists of two parts: 1-6th level Line Sequence code and 7-27th level Hilbert Curve code[23]. The 1st level of EMD subdivision divides the global region into 22 subareas which are encoded by a binary code of 5 bits. The 2nd level of EMD subdivision divides the

previously subdivided subarea into 4 meshes which are encoded by a binary code of 2 bits. In the same manner, the 3-6th level of binary code takes up 6 bits, 2bits, 2bits, and 4bits respectively. The 7-27th level of EMD subdivision divides the previous level of subdivided subareas into 4 meshes recursively, and every level takes up 2 bits respectively. Hence, the total length of SMC is 63 bits. SMC can identify a maximum fine-grained area of 4cm^2 , namely, the destination area of any customized granularity could be assigned a unique SMC.

In our design, we introduce the concept of the *domain* and the subordinate *subarea*. A domain refers to a country-scale region. A subarea refers to the area subdivided by the EMD subdivision method within a domain. Every domain and subarea has a unique SMC and that is the basis of implementing granularity-customizable geocast. In addition, the scale of a subarea is determined by the user requirements. But the destination area of the geocast may not be the same as the subarea. For instance, the destination area is usually a fine-grained area within a subarea when implementing the granularity-customizable geocast.

3.2 GeoIP Addressing

The mobility performance is another design goal of our solution. One of the main causes of the mobility problem in geocast is conflating identity and location by overloading a network identifier to represent both. In a geocast scenario, the communication endpoint is a 'geographical area' rather than a 'device'. Therefore, we introduce GeoIP[24] as the network layer identifier for packet forwarding. The structure of GeoIP is shown in Fig. 2.

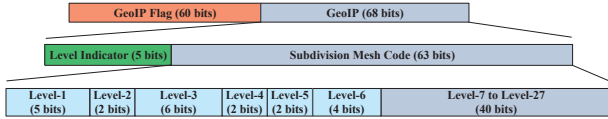


Figure 2: Structure of GeoIP

GeoIP is designed based on the SMC by adding 5 bits of binary code as the level indicator which is used to determine the level of SMC. To make it compatible with the current IP-base network, GeoIP is implemented with the IPv6 namespace. Hence, an additional 60 bits of GeoIP Flag are added to the GeoIP for completeness.

As the GeoIP is designed based on the SMC, a GeoIP could also uniquely represent a specific subdivided area. Hence, the GeoIP prefix of a domain is the same as the GeoIP prefix of its subordinate subarea. In our design, we assign the GeoIP of a subarea to the AP of that subarea to implement the granularity-customizable geocast. GeoIP is independent of the location of the end device, making the network layer forwarding insensitive to the changes in the receiver's location. Since the GeoIP Flag is fixed and the GeoIP only takes up 68 bits of IPv6 namespace, the cost of implementing GeoIP is relatively low.

3.3 GeoIP-based Packet Processing with P4

In our design, the granularity-customizable geocast is implemented at the edge of the network. When using the geocast service, the user will encapsulate the data packet with the GeoIP of an area. Then, the data packet will be sent to the AP of the subarea where

the destination area belongs, and then be further sent to the end devices of that subarea. To embed the GeoIP packet processing in the network, we use the P4-based SDN to implement the GeoIP-based forwarding and GeoIP packet receiving.

3.3.1 GeoIP Packet Forwarding Design. Every subarea has a wireless AP to implement geocast. Depending on the scale of the subarea, the wireless AP could be a base station, WIFI, or WLAN card. P4 is utilized as the programmable data plane of the AP to process the incoming GeoIP packets. The GeoIP packet forwarding design of the P4 pipeline is shown in Fig. 3.

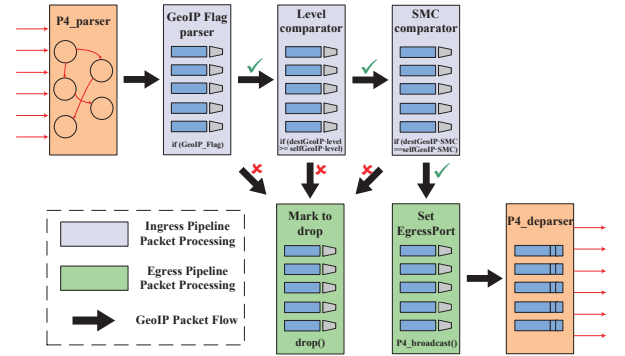


Figure 3: The GeoIP packet forwarding design of the P4 pipeline

When a GeoIP packet arrives at the P4 pipeline of the AP, the P4 parser will parse the packet header and then send the GeoIP packet to the ingress pipeline. The functions of the ingress pipeline are implemented based on P4 flow table matching. The GeoIP Flag parser will determine whether the packet is for geocast through the GeoIP Flag. If it is, the packet will be sent to the level comparator. Otherwise, the packet will be dropped. The level comparator will parse the level indicator of destination GeoIP. If the level of destination GeoIP is less than that of the AP's own GeoIP, the packet will be dropped. Otherwise, the packet will be sent to the SMC comparator. SMC comparator conducts a level-by-level comparison between the SMC of destination GeoIP and the SMC of AP's own GeoIP through the bitwise logical AND. The comparison is conducted for each level up to the level of AP's GeoIP. If each level of SMC is the same, the packet will be sent to the egress pipeline. Then the egress pipeline will conduct the *SetEgressPort* function and perform a P4 broadcast to send the GeoIP packet to the end devices connected to that AP. Otherwise, the packet will be dropped.

3.3.2 GeoIP Packet Receiving Design. As the GeoIP packets are forwarded through P4 broadcast, all the end devices connected to the AP would inevitably receive the GeoIP packet. However, only the users of the designated area are allowed to receive the GeoIP packets. To process the incoming GeoIP packets, a GeoIP packet receiving scheme is designed for the user client based on a dynamic flow table. The schematic diagram is shown in Fig. 4.

There are four modules on the control plane. The positioning application is a built-in module (say, a GPS) that could obtain the user's physical location and then write the latitude and longitude into the shared memory. The flow table manager is used to update the local flow table. If the flow table manager receives the

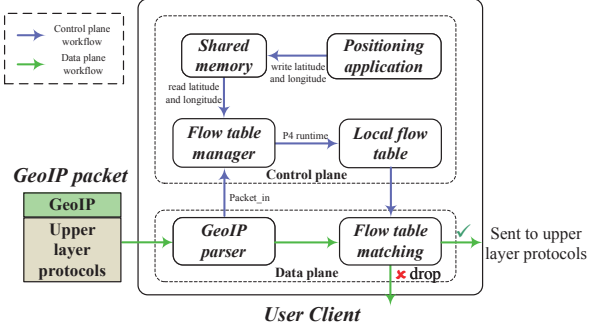


Figure 4: The schematic diagram of the GeoIP packet receiving scheme

packet_in packet, it will read the latitude and longitude from the shared memory and then convert it into a level-27 SMC through EMD coding. The flow table manager will use the converted SMC as the new matching field to update the local flow table through P4_runtime. The local flow table is used to determine whether the GeoIP packet should be received and it only has one entry. The data plane consists of two modules. When a GeoIP packet is received by the data plane, the GeoIP parser will send a packet_in packet to the control plane to trigger the flow table update. Then, the GeoIP parser will parse the level indicator of the destination GeoIP to obtain its level. The flow table matching module will perform a level-by-level matching between the SMC of the destination GeoIP and the entry of the local flow table. If the user is in the destination area, the prefix of the converted SMC will be the same as the SMC of the destination GeoIP, i.e., the entry of the local flow table will be hit. Then, the packet will be sent to upper-layer protocols for further processing. Otherwise, the packet will be dropped.

To support the existing Internet applications without making any changes to the protocol stacks or sockets, the GeoIP packet receiving scheme could be implemented by utilizing the programmable network interface card[25] or running a bmv2 daemon at the user client.

3.4 Geolocation Name Service

To make our design compatible with the current IP-based Internet, we borrow the idea of the Domain Name Service (DNS) system which resolves a human-readable Domain Name to an IP address. The GeoIP of the AP could be considered as the 'Domain Name'. Hence, a Geolocation Name Service (GNS) system is designed to resolve a GeoIP to an IP address. The GNS system is implemented as a control plane APP which consists of three kinds of mapping servers: the Domain GeoIP Server (DGS), the Root GeoIP Server (RGS), and the Local GeoIP Server (LGS).

- **DGS:** Each domain has a DGS which is assigned a public IP address. DGS maintains the mapping entries for the GeoIP of each subarea in that domain to the IP address of the local wireless AP in that subarea. When a subarea is added/deleted, the DGS shall add/remove the entries accordingly. If an entry is invalid in DGS, the DGS will inform the LGS to remove the invalid entry correspondingly.
- **RGS:** RGS records the mapping entries of each domain's GeoIP to the IP address of its DGS. Every RGS is assigned

a public IP address. Since the GeoIP prefix of a domain is the same as the GeoIP prefix of its subordinate subarea, RGS could tell the users which DGS they should query according to the prefix of destination GeoIP. When a domain is registered/removed in/from the network, the corresponding mapping entries should be added/removed to/from RGS.

- **LGS:** LGS is deployed in each subarea of a domain. It is responsible for recording the recently queried GeoIP-to-IP mapping entries to offload the query request in DGS and RGS. LGS could only be accessed by the users of the same subarea. When the entry exceeds its lifetime, it will be removed.

When a user uses the granularity-customizable geocast service, it will query the GNS first according to the destination GeoIP. The GNS APP will query the LGS. If an entry is hit, the LGS will return the result to the user. Then, the user will encapsulate a GeoIP packet with the IP address obtained from LGS and then send it to the Internet. Otherwise, the GNS APP will query the RGS to obtain the IP address of the DGS which it should query next. RGS will return the IP address of the specific DGS to the GNS APP. Then, the GNS APP will query the DGS to obtain the IP address of the destination AP and respond the result to the user. In addition, the GNS could be implemented by extending the existing DNS infrastructure to reduce the costs of deployment.

4 IMPLEMENTATION & EVALUATION

4.1 Design Implementation

To validate the proposed solution, bmv2 and WLAN cards are used as the data plane programmable AP to implement the proposed GeoIP packet forwarding scheme. We implement the GeoIP packet receiving scheme by utilizing P4Pi at the user client. The GNS system is implemented using MySQL daemon as virtual servers to maintain the GeoIP-to-IP mapping entries. ONOS is used for control plane management, and the GNS APP is implemented on the control plane as an ONOS APP.

Our prototype system consists of 40 bmv2 APs, 1 IP router, 4 virtual LGSs (to facilitate data measurement, LGSs are distributed by domain rather than by subarea as described in the previous section), 4 virtual DGSs, and 1 RGS. LGS is capacity-unconstrained to cache recently queried entries with a lifetime of 30 seconds. The testing area consists of 4 domains that are connected through an IP router. Domain_A and Domain_B are subdivided into 16 subareas with 16 bmv2 APs (each subarea has only one bmv2 AP). Domain_C and Domain_D are subdivided into 4 subareas with 4 bmv2 APs. 100 Raspberry PI devices (with positioning module) are randomly deployed into 40 subareas as the user client to send or receive the GeoIP packets. To simulate the user's movement, random noise is added to the positioning information of the Raspberry PI devices. The topology of the prototype system is shown in 5.

The bmv2 AP of a domain is deployed in a server with Ubuntu 18.04 system, Intel(R) Xeon(R) CPU E3-1230 V2@3.30GHz*8 4 Cores, Memory 8G. The LGS daemons, DGS daemons, and RGS daemons are respectively deployed in three servers with Ubuntu 18.04 system, Intel(R) Xeon(R) Silver CPU 4214 @2.20GHz*48 12 Cores, Memory 62.6G. The hardware information of the Raspberry PI is BCM2711B0 CPU@1.5GHz 4 Cores, 8GB, Raspberry Pi OS 5.10.17 32bit. The hardware of our prototype system is shown in 6.

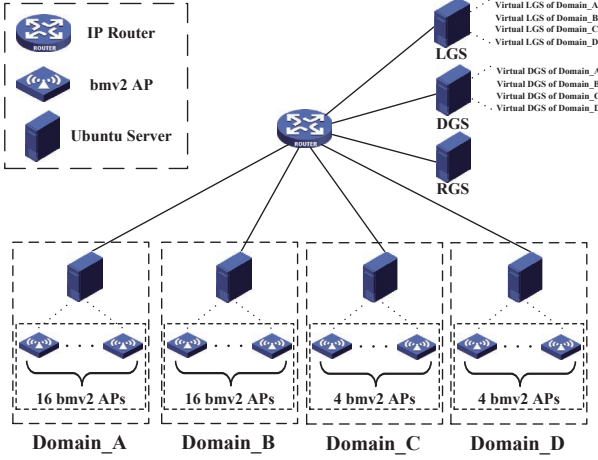


Figure 5: The topology of the prototype system

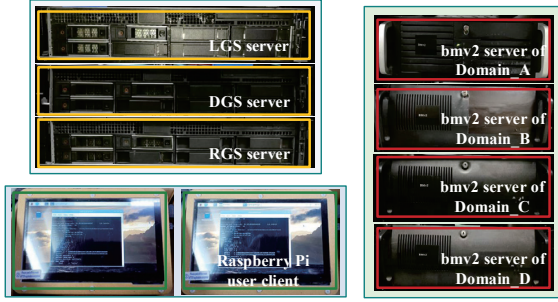


Figure 6: The hardware of the prototype system

4.2 Performance Evaluation

In this section, we will describe the metrics we are interested in, the scenarios that were used for the evaluation, and present the experiment results with analysis.

4.2.1 Evaluation Metrics. Our evaluation is focused on the following metrics:

- **The Response time for indexing different levels of GeoIP:** Since GeoIP consists of 27 levels of coding, one of the main overheads in our design is the response time for indexing different levels of GeoIP when querying entries in GNS.
- **The average time cost for processing different levels of GeoIP:** In our design, GeoIP packets are processed through GeoIP parsing, SMC comparing, and GeoIP-based flow table matching in bmv2. Therefore, the average time cost for processing different levels of GeoIP is an important network-layer performance. The average time cost for processing GeoIP packets includes the packet processing time of programmable AP and the processing time of the user client.
- **The number of queries to RGS/DGS:** Queries to RGS/DGS will be needed when implementing a cross-domain geocast. Traditional IP-based approaches for geocast are also based on the Geolocation-to-IP mapping services. Therefore, the number of queries to the mapping system is an essential performance for comparing the implementation cost of geocast.

4.2.2 Evaluation Scenario. We start the evaluation by running a packet sending daemon in each Raspberry Pi as the host of geocast. The packet sending daemon has an 80% probability of sending a GeoIP packet to a random destination area every second. The level of destination GeoIP ranges from 6 to 27. We set this packet sending probability to 80% as we refer to the flow intensity observed in our campus network. To compare with the IP-based approaches, the packet sending daemon will also conduct an IP-based geocast towards the same destination area after sending the GeoIP packet. The IP-based geocast is implemented using the service of the Geolocation-to-IP database. We use one IPinfoDB server to maintain the Geolocation-to-IP mapping entries of all the user clients. Four local Maxmind mapping servers are distributed by domain to maintain all the Geolocation-to-IP mapping entries for their administrative domains.

4.2.3 Results & Analysis. To evaluate the performance of our design, we ran nine tests, using level-3 to level-11 GeoIP as the index value of the mapping entry in turn. Then, the response time for indexing GeoIP-to-IP mapping entry in the GNS is measured ten times in each test as the data set. The result is shown in Fig. 7 by a box plot.

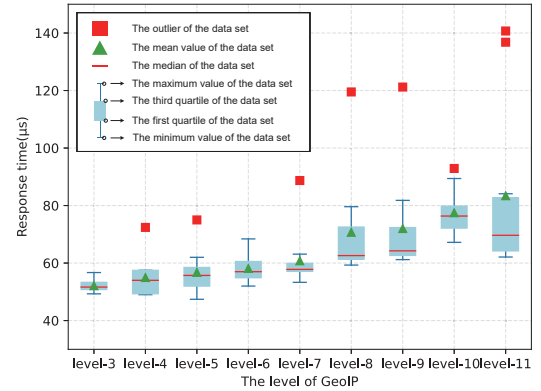


Figure 7: Response time for indexing different levels of GeoIP

The green triangle represents the mean value of the measured data set. The red square is the outlier. The red line of the box is the median of the measured data set. The lowest/highest blue line is the minimum/maximum of the data set excluding any outliers. The lower/upper boundary of the blue box represents the first/third quartile of the data set. As the level of GeoIP ranges from 3 to 11, the average response time ranges from 51μs to 82μs. Owing to the contention of system resources, the data set is not sufficient enough to fit the distribution curve of the average response time. But it is conjectural to conclude that the overhead for querying entries indexed by multi-level GeoIP will not exceed 200μs.

We measure the average time cost for processing level-6 to level-27 GeoIP in bmv2 and the result is shown in Fig. 8.

As the level of GeoIP ranges from 6 to 27, the processing time increases from 1311μs to 2403μs. The average time cost for processing different levels of GeoIP will not exceed 2.5ms. Considering that the processing speed mainly depends on the system hardware performance, the dedicated hardware switch or user client could be

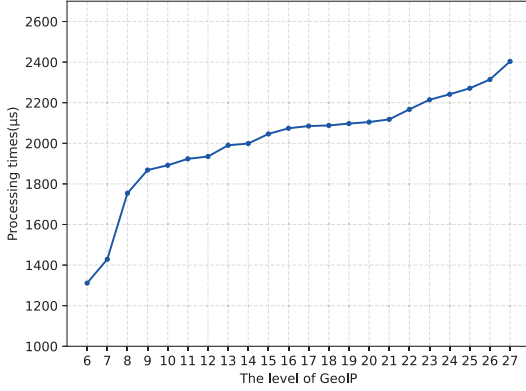


Figure 8: The average time cost for processing different levels of GeoIP in bmv2

used to reduce the GeoIP packet processing time. Since the geocast is implemented at the edge of the network, the time cost of less than 2.5ms is relatively low.

4.2.4 Comparison with the IP-based Approaches. One of the main overheads for IP-based approaches to implement granularity-customizable geocast is the number of mapping queries. Since the service of IPinfoDB and Maxmind are commonly used for location-based communication, we use the IPinfoDB and Maxmind servers as the contrast to RGS and DGSs respectively to evaluate the mapping querying cost. The number of queries to RGS, DGS, IPinfoDB, and Maxmind is measured every second and shown in Fig. 9 and Fig. 10.

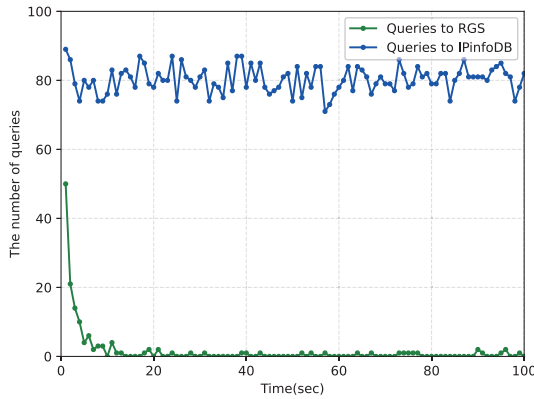


Figure 9: The number of queries to RGS vs IPinfoDB

In the first five seconds of the experiment, the number of queries to RGS is 50, 21, 14, 10, and 4, which are much smaller than that of the IPinfoDB. That is because the query to RGS will not be necessary when the host of geocast and the packet receivers are present in the same domain. While the IPinfoDB server needs to maintain the Geolocation-to-IP mapping entries of all the user clients. When the host of geocast and the packet receivers are present in different domains, additional queries to the DGS of the destination area are needed. While local Maxmind servers could maintain all the required entries for their administrative domain. As a result, in

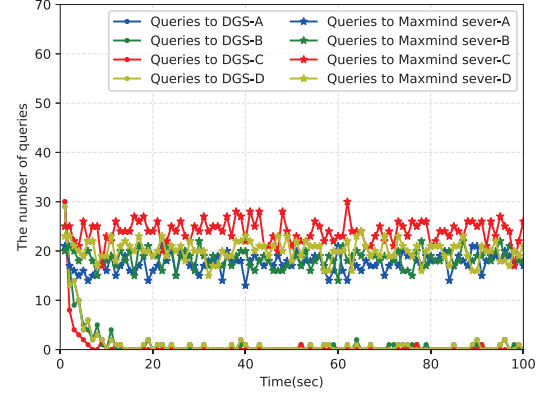


Figure 10: The number of queries to DGS vs Maxmind

the first second of the experiment, the average number of queries to the DGSs is 28.75 which is a little larger than that of the local Maxmind servers of 22.25. As the experiment continues, necessary entries will be cached into LGSs. Thus, the query requests to RGS and DGSs will be offloaded to the LGSs gradually. The query to RGS and DGSs is only needed when some entries in LGSs are invalid after exceeding their lifetime.

As for the LGS, the number of queries to LGSs is only relevant to the number of hosts. To offload the query request to DGSs and RGS, the capacity of LGS should be sufficient enough to cache potentially queried entries. The lifetime of the entry cached by LGS should be prolonged if no subarea is registered or removed.

5 CONCLUSION & FUTURE WORK

In this paper, we present a GeoIP-based network layer solution to enable granularity-customizable geocast for the current Internet. The EMD method is used to implement the GeoIP-based global subdivision and addressing. Then, we use P4-based SDN to embed the GeoIP packet processing scheme at the edge of the network. Moreover, a Geolocation Name Service is designed to make our solution compatible with the current IP-based Internet. To validate and evaluate our work, we use the bmv2, ONOS, and MySQL software to implement the proposed design in a prototype system. Experiment results show that our solution is feasible to provide granularity-customizable geocast at a relatively low cost.

For future work, the GeoIP-based addressing provides a foreseeable benefit for geolocation-oriented services. However, the explicit geolocation semantics of the GeoIP makes it easy to expose the important geolocation information, making the network vulnerable to geolocation-specific attacks. Therefore, we are planning to modify the GeoIP packet receiving scheme by designing a GeoIP-based data encryption mechanism. Meanwhile, we will focus on how the packet should be forwarded when the destination area spans across multiple subareas or when the destination area is an irregular rectangle.

6 ACKNOWLEDGMENTS

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